

FLOATING POINT REPRESENTATION

PRACTICE SHEET • EASY / FUNDAMENTALS LEVEL

ResourceOS

1. How many total bits are used in the standard IEEE 754 single-precision floating-point format?

- (A) 16 bits
- (B) 32 bits
- (C) 64 bits
- (D) 128 bits

2. In IEEE 754 floating-point format, what does a sign bit value of 0 represent?

- (A) Negative number
- (B) Positive number
- (C) Zero exponent
- (D) Overflow condition

3. How many bits are allocated for the exponent field in IEEE 754 single-precision (32-bit) format?

- (A) 8 bits
- (B) 11 bits
- (C) 23 bits
- (D) 52 bits

4. How many bits are allocated for the mantissa (fraction) field in IEEE 754 single-precision (32-bit) format?

- (A) 8 bits
- (B) 16 bits
- (C) 23 bits
- (D) 31 bits

5. What is the bias value added to the actual exponent in IEEE 754 single-precision format?

- (A) 63
- (B) 127
- (C) 255
- (D) 1023

6. How many total bits are used in the IEEE 754 double-precision floating-point format?

- (A) 32 bits
- (B) 48 bits
- (C) 64 bits
- (D) 80 bits

7. What is the exponent bias value used for IEEE 754 double-precision (64-bit) floating-point format?

- (A) 127
- (B) 255
- (C) 511
- (D) 1023

8. In normalized floating-point numbers, what is the implicit leading bit before the binary point?

- (A) 0
- (B) 1
- (C) Depends on sign bit
- (D) Variable bit

9. What is the binary representation of the fractional decimal number 0.5?

- (A) 0.1_2
- (B) 0.01_2
- (C) 0.11_2
- (D) 0.001_2

10. Convert the decimal number 0.25 to binary fractional form.

- (A) 0.1_2 (B) 0.01_2
(C) 0.001_2 (D) 0.11_2

11. What is the normalized binary scientific notation representation for decimal 6.0?

- (A) $1.1_2 \times 2^2$ (B) $11.0_2 \times 2^1$
(C) $0.11_2 \times 2^3$ (D) $1.1_2 \times 2^1$

12. If an actual exponent in single precision is 3, what is the stored (biased) exponent field value in decimal?

- (A) 130 (B) 124
(C) 127 (D) 131

13. If the stored 8-bit exponent field in IEEE 754 single-precision is 10000000_2 (128 in decimal), what is the actual exponent?

- (A) +1 (B) 0
(C) -1 (D) +128

14. What sign bit is stored for the number -8.5 in floating-point format?

- (A) 1 (B) 0
(C) -1 (D) Depends on exponent

15. What is the binary fractional value of 0.75?

- (A) 0.11_2 (B) 0.101_2
(C) 0.011_2 (D) 0.111_2

16. In floating-point arithmetic, why is exponent biasing used instead of standard 2's complement for exponents?

- (A) To simplify floating-point comparison using integer magnitude operations (B) To reduce memory consumption
(C) To eliminate fractional bits (D) To allow faster addition of mantissas

17. In IEEE 754 single precision, what value does an exponent field of all 1s (255) with a mantissa of all 0s represent?

- (A) Infinity (∞) (B) Zero
(C) Not a Number (NaN) (D) Denormalized number

18. In IEEE 754 single precision, what value does an exponent field of all 1s (255) with a non-zero mantissa represent?

- (A) Not a Number (NaN) (B) Infinity
(C) Underflow (D) Overflow limit

19. Convert decimal 1.0 into normalized binary floating-point form ($1.M \times 2^E$). What is its actual exponent?

- (A) 0 (B) 1
(C) 127 (D) -127

20. What is the stored 8-bit exponent for decimal 1.0 in IEEE 754 single precision?

- (A) 01111111 (127) (B) 10000000 (128)
(C) 00000000 (0) (D) 11111111 (255)

21. In IEEE 754 format, what are numbers with an exponent field of all 0s and a non-zero mantissa called?

- (A) Denormalized (or Subnormal) numbers (B) Normalized numbers
(C) Overflow numbers (D) NaN values

22. Which component of a floating-point number determines its overall magnitude and scale?

- (A) Exponent (B) Mantissa
(C) Sign bit (D) Parity bit

23. Which component of a floating-point number determines its precision?

- (A) Mantissa (Fraction) (B) Exponent
(C) Sign bit (D) Bias constant

24. Convert binary 10.11_2 into normalized scientific notation.

- (A) $1.011_2 \times 2^1$ (B) $1.011_2 \times 2^2$
(C) $10.11_2 \times 2^0$ (D) $0.1011_2 \times 2^2$

25. What is the mantissa field bit pattern stored for decimal 1.5 in single precision?

- (A) 100000000000000000000000 (B) 010000000000000000000000
(C) 110000000000000000000000 (D) 000000000000000000000000

ANSWER KEY

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	(B)	6	(C)	11	(A)	16	(A)	21	(A)
2	(B)	7	(D)	12	(A)	17	(A)	22	(A)
3	(A)	8	(B)	13	(A)	18	(A)	23	(A)
4	(C)	9	(A)	14	(A)	19	(A)	24	(A)
5	(B)	10	(B)	15	(A)	20	(A)	25	(A)